

Cell Cycle and Checkpoints

What each checkpoint is actually testing, and what specifically comes out of the cell when one of them is missing.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/cell-cycle/

1 Before you open anything

Answer from what you already know. Commit to something on paper — the value of this page comes from having a prediction to compare against, not from being right first.

PREDICTION

In the model you are about to open, a cell surrounded by **100% growth factors** has a G_1 of about 11 h and a total cycle of 24 h (G_1 11 h, S 8 h, G_2 4 h, M 1 h). The growth-factor level outside the cell is then lowered to **45%**. Circle one:

a. The cell never divides again. **b.** It divides on the same 24 h schedule. **c.** It divides, but G_1 alone is longer — roughly _____ h. **d.** It divides, but every phase is stretched by the same factor.

Write one sentence saying what reasoning got you there.

2 One undisturbed cycle, timed

Open the demonstration. The panel *This cell right now* reports where the cell is; the line under **Position in the cycle** gives the hours elapsed in the current cycle and the length that phase normally takes.

1. Under **Presets**, press **Healthy cell**. This resets the run and clears every fault.
2. Set the **Speed** slider to **2.0 h/s**.
3. Watch the large value under **Position in the cycle**. The instant it changes letter, press **Pause** and read the line beneath it ("___ h into this cycle"). Record, then press **Run** again.
4. Do this for the change into S, into G_2 and into M. Drop the Speed to **0.5 h/s** once you reach G_2 , because M is short.
5. Also copy the normal length the readout quotes for each phase ("normal ___ h for S").

Phase	"h into this cycle" when this phase begins	Length of the phase (next start - this start)	"normal ___ h" quoted by the readout
G ₁	0.0		
S			
G ₂			
M			

a. Watch card 1 · G₁ / **restriction point** during G₁. At what reading of "h into this cycle" does its state change from **TESTING** to **PASSED — past the restriction point**?

b. Copy the sentence that card then displays. What does it claim stops mattering to this cell from that moment on?

c. What does **DNA replicated** read at the moment the phase changes from S to G₂? What does **Kinetochores attached** read just before anaphase starts?

d. Let the division finish. From *What the last division actually produced*, record all three: **Genome copied while it was damaged** _____, **DNA replicated at the moment mitosis began** _____, **Daughter chromosome numbers** _____. The note underneath ends with the cycle length — record that too: _____ h.

Nothing in this model runs on a clock. Each transition happens when a cyclin–CDK activity crosses a threshold, which is why the cyclin plot — not the wheel — is the mechanism.

3 How much of the cycle the outside world controls

One control, varied systematically. You are changing only the mitogen signal arriving at the cell surface, and measuring what that does to the length of G_1 .

1. Press **Healthy cell** to clear everything.
2. Set **Growth factors outside** to the value in the first column (the slider runs 0–100% in steps of 1%).
3. Press **Reset all** so the cell restarts in G_1 with that signal from the beginning.
4. Run at **4.0 h/s** and pause near **5.0 h into this cycle**. Card 1 • G_1 / **restriction point** is reading TESTING and gives the cyclin E–CDK2 activity as a percentage; copy it into the third column, then resume.
5. For the bottom three rows raise the Speed to **16.0 h/s** after taking that reading, since those cells sit in G_1 far longer.
6. Pause the instant **Position in the cycle** changes to S and record the hours. For the last row: if the reading passes **300 h** and the cell is still in G_1 , write “does not enter S” and move on.

Growth factors outside	Length of G_1 (h)	Cyclin E–CDK2 activity shown on card 1 at 5 h (% of the level that starts S)
100%		
80%		
60%		
50%		
45%		
40%		
30%		

- a. For the 60% row only, let the cell run on into G_2 and measure the length of **S phase**. Compare it with the S phase you measured in Part 2 at 100%. Same, longer or shorter?

b. State the rule in your own words: what does lowering the growth-factor level do to G_1 , and what does it do to S, G_2 and M? Is the relationship proportional?

c. Somewhere between two of your rows the cell stops entering the cycle altogether. Read the hint printed under the **Growth factors outside** slider: what level does it name as the cut-off, what is the resting state called, and which protein is never phosphorylated below that level?

d. Go back to your Part 1 prediction. Which option was right? If yours was not, name the specific assumption that turned out to be false.

4 Reading a daughter cell backwards

A free-response question rarely hands you the fault. It hands you the daughter cell and asks which control failed. Build the evidence table first, then work in that direction.

1. Press each preset in turn: **Healthy cell**, **p53 lost**, **G₂ lost**, **Spindle checkpoint lost**.
2. Each press resets the run. Wait until **Divisions completed** reaches **1**, then read the three values in *What the last division actually produced*.
3. Speed **4.0 h/s** is comfortable for this.

Preset	Genome copied while it was damaged	DNA replicated at the moment mitosis began	Daughter chromosome numbers
Healthy cell			
p53 lost			
G ₂ lost			
Spindle checkpoint lost			

a. Three of your four rows report a normal chromosome count. Only one of those three is a healthy division. Explain, in one or two sentences, why a karyotype is not a general test for whether a division went well.

b. Work backwards. For each described daughter cell, name the single checkpoint whose loss produced it.

6 chromosomes, every gene present, but thousands of base changes that were not in the parent:

7 chromosomes; every gene present is intact and correctly sequenced:

6 chromosomes, but whole stretches of several chromosomes are absent:

A shortcut worth testing rather than trusting: *if the G₁ checkpoint misses damaged DNA, the G₂ checkpoint will catch it later*. The next two questions put that to the demonstration.

c. Select **p53 lost** again. The **G₂ checkpoint working (ATR–Chk1 present)** box is still ticked, and the cell entered S carrying heavy damage. Watch card 2 · **G₂ checkpoint** through S and G₂. Does it ever read **HOLDING**? Record what **Damage unrepaired** reads as the cell enters G₂.

d. Explain that result. Where did the damage go during S phase, and why can no downstream checkpoint detect what it became?

e. Now the restriction point, tested directly. Press **Healthy cell**, set Speed to **0.5 h/s**, and pause when the readout is near **5.0 h into this cycle**. Drag **Growth factors outside** to **0%**, run on, and record what the cell does over the next 20 h. Then press **Reset all** and repeat, this time removing the growth factors at about **9.0 h**.

Removed at 5.0 h:

Removed at 9.0 h:

f. Both cells lost the same signal. Account for the difference using cyclin D, p27 and Rb.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Then reopen it and check yourself, and where you were wrong, write down which step went off.

1. Use the undisturbed timings: G_1 11 h, S 8 h, G_2 4 h, M 1 h. A cell is 20.0 h into its cycle. Name the phase and say how far into that phase it is. This model cell has 6 chromosomes: at that moment, how many chromatids does each chromosome have, and how many DNA molecules are in the cell in total?

2. The restriction point falls about 7.5 h into G_1 . Growth factors are stripped from two identical cells: cell A at 5.0 h, cell B at 9.0 h. Predict what each does, and explain both outcomes in terms of cyclin D, p27 and Rb. Then state which of the two would be predicted incorrectly by the rule “no growth factor, no division.”

3. A division produces one daughter with 7 chromosomes and one with 5. Every gene present is intact and correctly sequenced, and both daughters have a complete copy of every chromosome they received. Name the checkpoint that failed and the exact point in mitosis at which it acts. Then explain why neither of the other two checkpoint failures could have produced this particular result.

4. At 100% growth factors the cycle is 24 h with an 11 h G_1 . At 45% growth factors G_1 lasts about 27 h and nothing else changes. Calculate the total cycle length at 45%, and calculate the percentage of the cycle spent in G_1 at each of the two signal levels. State what those two percentages imply about which part of the cycle a tissue actually regulates.

6 On your own

Nothing here is graded, and the demonstration does considerably more than this worksheet uses.

- Set **Replication stress** to 100% but leave the G_2 checkpoint ticked. Time S phase and read card 2 while it holds. Then untick the checkpoint and run it again. Both cells have the same problem; only one of them survives it intact.
 - Set **Spindle problems** to 55% with the spindle assembly checkpoint left *working*. The cell parks at metaphase instead of dividing. That arrest is the mechanism by which taxol and the vinca alkaloids kill dividing cells — check the honest-limits panel for what a real cell does after several hours of it.
 - Tick **Growth-factor independent** with growth factors at 0% and run several divisions, watching *Damage accumulated in this lineage*. Then untick the G_1 checkpoint, set **DNA damage each cycle** to 70%, and run several more. Note what the panel says about how many independent defects are present, and what it takes before the word cancer is warranted. The **Several hits at once** preset is the same idea taken further.
 - Watch the cyclin plot through one mitosis and find the moment cyclin B collapses. Name the event, name the enzyme complex responsible, and explain why the cell leaves mitosis because of a destruction rather than because of an elapsed time.
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